UNDERSTANDING AND PREDICTING VÉLIB' BIKE AVAILABILITY IN PARIS USING TEMPORAL AND WEATHER DATA

A PREPRINT

Emirhan Faruk SAGIR
Sorbonne Université
emirhan.sagir@etu.sorbonne-universite.fr

Arafat ISLAM
Sorbonne Université
arafat.islam@etu.sorbonne-universite.fr

ABSTRACT

Bike-sharing systems play a crucial role in modern urban transportation. Their efficiency, however, strongly depends on the real-time availability of bikes at stations. In this project, we study whether temporal, spatial, and weather-related factors can be leveraged to predict the number of available Vélib' bikes at a given station in Paris. We collect large-scale real-time data from public APIs, design a complete data processing pipeline, and evaluate several regression models. Beyond prediction performance, we analyze the actual influence of weather on bike usage and discuss the practical and methodological limitations encountered when working with real-world urban data.

1 Introduction

Bike-sharing services have become an essential component of sustainable urban mobility. In large cities such as Paris, Vélib' provides an alternative to private vehicles and public transportation, reducing congestion and environmental impact. Despite its popularity, users frequently encounter empty or saturated stations, which negatively affects service reliability.

Predicting bike availability is therefore of both operational and societal interest. From an operator's perspective, accurate predictions may support better rebalancing strategies. From a scientific perspective, this task raises interesting questions regarding the influence of environmental factors versus habitual human behavior. This project follows the full data science pipeline taught in the DALAS course, from data collection to modeling and interpretation.

2 Problem Statement and Objectives

The central research question addressed in this project is the following:

> To what extent can short-term Vélib' bike availability at a given station be predicted using historical usage patterns, temporal features, and weather-related information?

This problem is formulated as a supervised regression task, where the target variable is the number of available bikes at a station at a given timestamp. The study focuses on short prediction horizons, where recent historical information may play a dominant role.

The main objectives of this project are to:

- Build a large-scale dataset combining station-level bike availability data with meteorological information.
- Clean, preprocess, and merge heterogeneous data sources while avoiding temporal leakage.
- Engineer temporal, contextual, and autoregressive features relevant to short-term prediction.
- Cluster stations based on long-term usage patterns to account for spatial and behavioral heterogeneity.
- Train and compare regression models of varying complexity.

- Analyze the relative contribution of historical usage, temporal structure, and weather variables to predictive performance.

In summary, this project aims to predict short-term bike availability at Vélib' stations in Paris using historical usage, temporal features, and weather data. A large-scale dataset was collected over nearly two months, carefully cleaned, and enriched with contextual information. The study evaluates whether weather variables provide predictive value beyond short-term usage history, and shows that recent availability dominates prediction accuracy.

3 Data Collection

3.1 Data Source

We collected data from two complementary public sources summarized in Table 1. Station-level bike availability data were obtained from the Vélib' Open Data platform [1], which provides real-time snapshots of the entire bike-sharing network in Paris. Weather observations were retrieved from the Open-Meteo API [2], selected after evaluating several alternatives due to its free access and sufficient temporal resolution, despite its rate-limiting constraints and non-commercial usage policy.

Table 1: Summary of Data Sources Used in the Study

Data Source	Variables	Temporal Resolution
Vélib' Open Data [1]	Station identifier, name, latitude, longitude, station capacity, available bikes (total, mechanical, electric), available docks	15–20 minutes
Weather Data [2]	Temperature (2 m), wind speed (10 m), wind direction (10 m), weather code, day/night indicator	15–20 minutes

To ensure continuous data acquisition over an extended period, the collection pipeline was deployed on a Raspberry Pi 4B and executed uninterrupted for approximately 48–50 days, between the dates 04/11/2025 and 23/12/2025. The data acquisition and integration process followed the procedure described in Algorithm 1.

Algorithm 1 Periodic Data Collection and Integration Pipeline

while data collection is running **do**
  Fetch Vélib' station snapshot V (stations, capacity, availability)
  Save V to JSON
  Define weather query points (arrondissements and communes)
  for all weather points p **do**
    Fetch weather observations W_p at point p
    Wait 0.5 s ▷ rate limiting
  end for
  $W \leftarrow \bigcup_p W_p$
  Save W to JSON
  for all stations $s \in V$ **do**
    $p^*(s) \leftarrow \arg\min_p d(s, p)$ ▷ nearest weather point
  end for
  $D \leftarrow \{(s, W_{p^*(s)}) : s \in V\}$ ▷ merge station and weather
  Save merged dataset D to JSON
  Wait $U(15, 20)$ min ▷ random delay
end while

At each iteration, a snapshot of Vélib' station data was collected and paired with contemporaneous weather observations. Each station was assigned to its nearest weather observation point based on Euclidean distance, after which both data sources were merged into a single snapshot. The pipeline was executed periodically with randomized intervals between 15 and 20 minutes to comply with API rate limits and to avoid temporal synchronization artifacts.

3.2 Data Aggregation and Integration

The Vélib' station data and weather observations originate from independent data sources with different spatial resolutions. To integrate these sources into a single, coherent dataset, we adopted a snapshot-based spatial matching strategy.

Weather observations were queried at a fixed set of 86 geographic reference points, corresponding to the 20 Paris arrondissements and 66 surrounding communes. For each data collection cycle, Vélib' station snapshots and weather observations were treated as temporally aligned. Each station was then associated with the most representative weather observation based on spatial proximity.

Specifically, let s denote a Vélib' station with coordinates (x_s, y_s), and let p denote a weather observation point with coordinates (x_p, y_p). The spatial distance between a station and a weather point was computed using the Euclidean distance

$$d(s,p) = \sqrt{(x_s - x_p)^2 + (y_s - y_p)^2}, \tag{1}$$

where $d(s,p)$ represents the geographic distance between station s and weather point p. Each station s was assigned to the weather point $p^*(s)$ minimizing this distance, defined as

$$p^*(s) = \arg\min_p \; d(s,p). \tag{2}$$

The weather features associated with $p^*(s)$ were then attached to the corresponding station snapshot. This procedure ensures that each station observation is augmented with locally representative weather conditions while preserving the original temporal structure of the data.

Figure 1 provides a spatial overview of the Vélib' stations and illustrates, through a representative example, the nearest-neighbor assignment of a station to its closest weather observation point according to Equation 2.

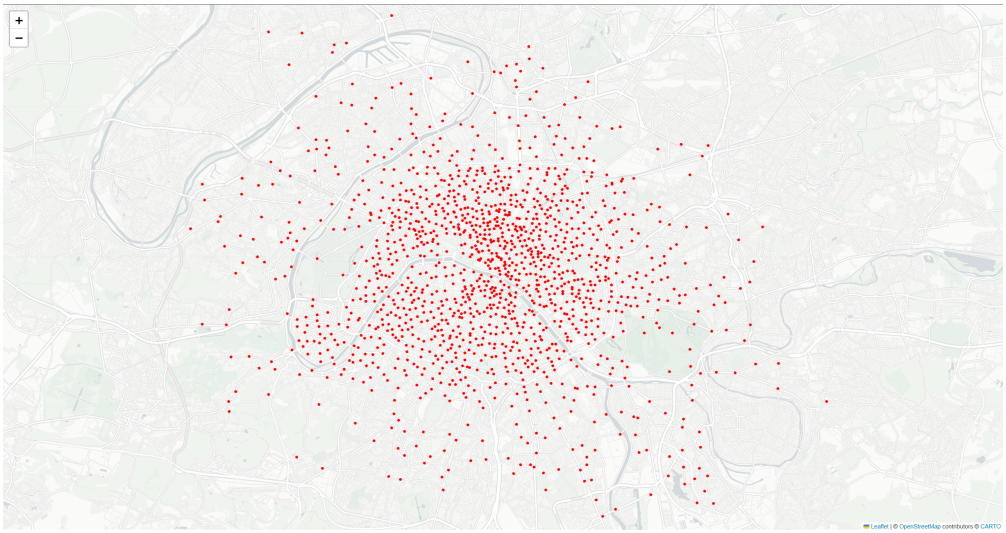


Figure 1: Spatial distribution of Vélib' stations in Paris.

3.3 Dataset Overview

Table 2 summarizes the dataset at the main acquisition and integration stages, from the raw JSON snapshots to the merged tabular version used for subsequent preprocessing and modeling. The resulting dataset contains station-level observations enriched with contextual weather information.

A detailed definition of the available variables, including their types and semantic meaning, is provided in Table 3.

Table 2: Dataset Overview

Stage	Format	Rows	Features	Size
Raw data collection	JSON (4048 files)	1505 stations per file	23	5.83 GB
Merged dataset	CSV	6,070,326	23	1.16 GB

Table 3: Description of Raw Dataset Features

Feature	Type	Description
station_name	Categorical(string)	Vélib' station name.
stationcode	Categorical(int)	Unique Vélib' station identifier.
latitude	Numeric(float)	Station latitude (WGS84).
longitude	Numeric(float)	Station longitude (WGS84).
capacity	Numeric(int)	Total number of docks at the station.
numbikesavailable	Numeric(int)	Number of bikes currently available.
numdocksavailable	Numeric(int)	Number of empty docks available.
ebike	Numeric(int)	Number of electric bikes available.
mechanical	Numeric(int)	Number of mechanical bikes available.
nom_arrondissement_communes	Categorical(string)	Commune or arrondissement of the station.
code_insee_commune	Categorical(int)	INSEE code of the corresponding commune.
duedate	Temporal(datetime)	Timestamp of the Vélib' snapshot.
is_installed	Binary(int)	Indicator of station installation status.
is_renting	Binary(int)	Indicator of whether renting is enabled.
is_returning	Binary(int)	Indicator of whether returns are enabled.
weather_time	Temporal(datetime)	Timestamp of the associated weather observation.
weather_interval	Temporal(datetime)	Time interval of the weather observation.
weather_temperature_2m	Numeric(float)	Air temperature at 2 m above ground (°C).
weather_wind_speed_10m	Numeric(float)	Wind speed at 10 m above ground.
weather_wind_direction_10m	Numeric(float)	Wind direction at 10 m above ground.
weather_weather_code	Numeric(int)	Encoded weather condition indicator.
weather_is_day	Binary(int)	Indicator distinguishing day and night conditions.

Table 4: Dataset Evolution Across Major Processing Stages

Processing Stage	Format	Rows	Features	Size
Raw data collection	JSON (4,048 files)	1,505 stations per snapshot	23	5.83 GB
Merged raw dataset	CSV	6,070,326	23	1.16 GB
After capacity anomaly removal	DataFrame	5,940,155	23	$\sim$1.1 GB
After duplicate aggregation	DataFrame	5,876,315	17	739.7 MB
After corrupted station and row filtering	DataFrame	5,644,165	17	753.58 MB
After temporal feature engineering	DataFrame	5,888,930	17	764 MB
Final preprocessed dataset	CSV	6,705,169	20	1.02 GB

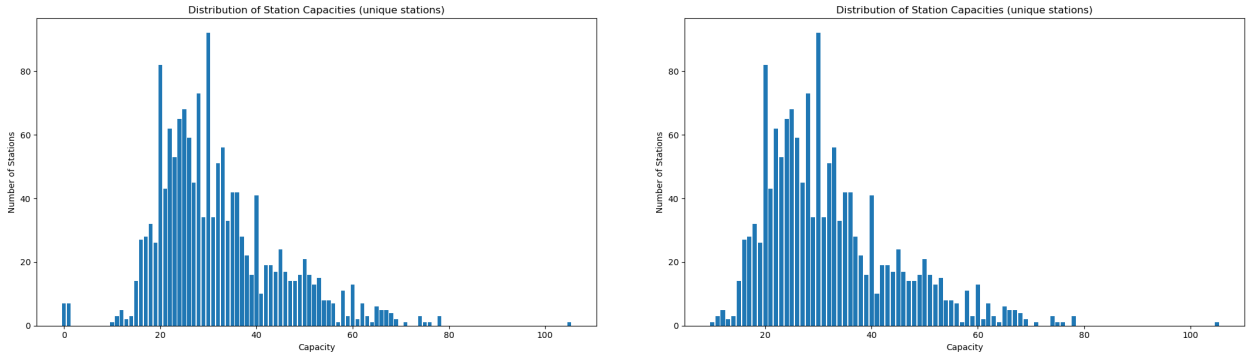
After exceeding the request quota during early experiments, we redesigned our collection strategy to drastically reduce the number of API calls. Instead of querying weather data for each station, we selected representative weather points at the commune or arrondissement level. Each Vélib' station was then assigned to its nearest weather point based on geographic distance. This way, the number of requests was reduced from 1500 to 86(20 arrondissements and 66 communes) per 15-20 minutes.

4 Data Quality and Preprocessing

4.1 Capacity Anomaly Detection and Removal

During exploratory analysis of station metadata, we inspected the distribution of station capacities to identify invalid or non-operational stations. The histogram of unique station capacities (Figure 2a) revealed a small number of stations with reported capacities equal to 0 or 1, which is inconsistent with a functional bike-sharing station. Such values likely correspond to stations that were not yet operational at the time of collection, temporarily unavailable, or affected by reporting errors.

To ensure internal consistency and avoid introducing noise into downstream modeling, we removed stations with capacity ≤ 1 and also filtered out any remaining rows violating the constraint `capacity` > 1. After applying this rule, a total of 28 stations were excluded from the dataset. The resulting capacity distribution after cleaning is shown in Figure 2b.



(a) Before capacity filtering.

(b) After removing stations with capacity ≤ 1.

Figure 2: Distribution of station capacities before and after removing invalid capacities.

4.2 Corrupted Usage Records (Bike Count Exceeding Capacity)

After removing stations with invalid capacities, we examined the consistency between station capacity and the number of available bikes. Figure 3 shows the relationship between these two variables across all remaining stations. While most observations satisfy the physical constraint that the number of available bikes does not exceed station capacity, a non-negligible number of records violate this condition, indicating partially corrupted data.

Rather than discarding all corrupted observations indiscriminately, we analyzed the prevalence of such violations on a per-station basis. For each station, we computed the ratio of corrupted rows—defined as observations where the number of available bikes exceeded the reported capacity—to the total number of observations. This analysis revealed two distinct regimes.

A small subset of stations exhibited a high proportion of corrupted records, with 9 stations exceeding a 10% corruption threshold, including two stations for which more than half of the observations were invalid. These stations were entirely removed from the dataset, as retaining them would require extensive interpolation and risk introducing artificial patterns.

In contrast, 22 stations exhibited a low proportion of corrupted rows (below 10%). For these stations, only the corrupted observations were removed, while the remaining data were preserved. This strategy allowed us to retain informative stations while eliminating clear data inconsistencies. A summary of station-level corruption statistics is provided in Table 5.

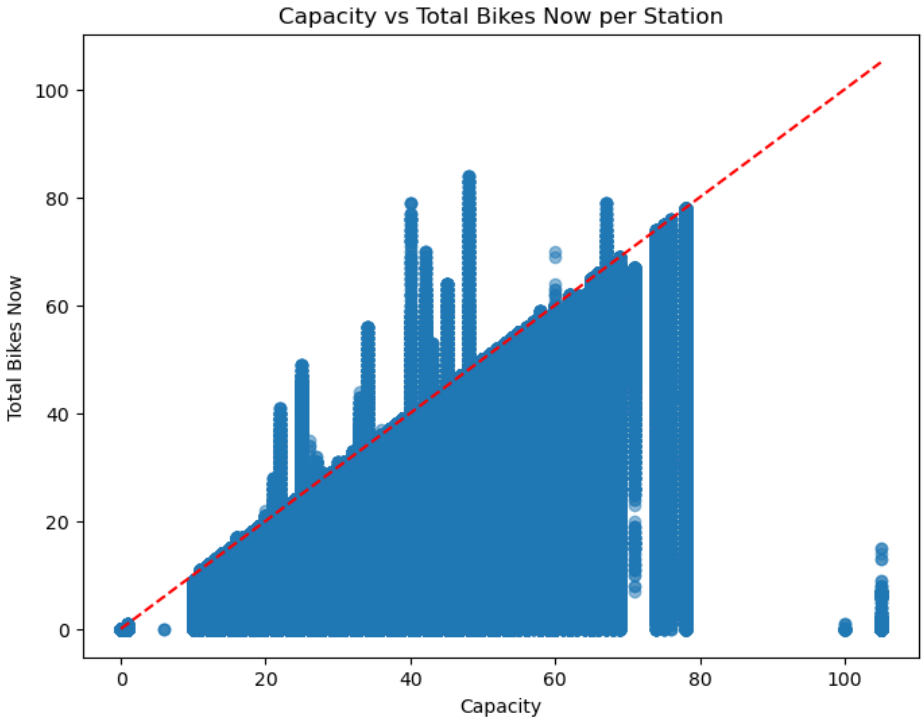


Figure 3: Relationship between station capacity and number of available bikes. Points above the diagonal indicate corrupted observations where bike counts exceed station capacity.

Table 5: Stations with the Highest Proportion of Corrupted Observations

Station Code	Total Rows	Corrupted Rows	Corruption Ratio
21302	3970	2319	0.58
42004	4014	2165	0.54
13123	4014	1713	0.43
15056	2543	746	0.29
12105	2579	569	0.22

4.3 Duplicate Station–Timestamp Resolution

Despite the overall consistency of the collected data, temporal visualizations of station activity revealed unexpected gaps in station timelines. Although the total number of rows was close to the expected count, timestamp heatmaps showed a large number of missing time points, even for stations with otherwise dense observations.

This behavior was traced to the presence of duplicate `station_code–timestamp` pairs. These duplicates arose primarily from timestamp alignment to fixed 15-minute intervals and from repeated use of the same weather observation timestamps during periods of delayed weather updates.

To resolve this issue, duplicate station–timestamp pairs were aggregated using feature-specific rules designed to preserve the most representative information. Station metadata and temporal indicators were retained using the first observed value, while usage and weather variables were averaged. This aggregation restored temporal consistency without discarding valid information.

4.4 Detection and Removal of Low-Quality Stations

Following data cleaning and aggregation, we evaluated the temporal coverage of each station. A histogram of the number of observations per station revealed a highly skewed distribution, with a small subset of stations exhibiting extremely sparse data.

Stations with insufficient data coverage were identified using a quantile-based threshold, set at the 5% quantile of the station-level row count distribution. In addition, stations exhibiting near-constant or stagnant behavior were detected using statistical criteria based on the mean and standard deviation of bike availability. Stations with negligible variation or undefined statistics were considered non-informative and removed.

This combined strategy resulted in the removal of 73 low-quality stations, leaving 1,417 stations suitable for modeling.

4.5 Dataset State Prior to Timeline Completion

After removing low-quality stations, the dataset was analyzed to assess the impact of preprocessing relative to the initial collection. The proportion of removed rows and stations remained modest, particularly when considering only valid observations. This confirmed that the cleaning pipeline primarily eliminated unreliable data while preserving the vast majority of informative records.

4.6 Temporal Completion of Station Timelines

To ensure consistent temporal structure across stations, complete station timelines were constructed by identifying the earliest and latest observation timestamps for each station and inserting missing intermediate time points. Missing values were left unfilled at this stage to prevent information leakage.

This process increased the total number of rows by introducing explicit placeholders for missing observations, enabling consistent time-series modeling in later stages.

4.7 Missing Value Imputation

Missing values were filled using feature-specific strategies. Static spatial attributes, including station coordinates and capacity, were imputed using forward-fill methods. Weather variables were filled similarly, reflecting the gradual evolution of meteorological conditions.

Bike availability variables required more careful treatment and were first imputed using time-based interpolation, followed by forward-fill for boundary values. Temporal features were reconstructed after imputation to ensure consistency.

4.8 Lag Feature Construction

To capture short-term temporal dependencies in bike availability, lagged features were introduced by shifting the bike availability time series by one and two time steps. These lag features provide models with access to recent station history and improve short-term predictive performance.

4.9 Final Dataset Assembly

After completing all preprocessing and feature construction steps, the independently processed dataset splits were merged into a single consolidated dataset and saved for model training and evaluation. A final analysis of dataset size, station coverage, and split composition is provided in the dataset evolution summary.

5 Data Collection Challenges

5.1 Continuous Data Collection Challenge

We had to find a way to collect the Vélib' and weather data nonstop, as the data continuity is vital. We have decided to use a Raspberry Pi 4B to let the code run independently. We had to collect the data from Raspberry Pi using scp every 1-3 weeks due to storage limitations on the device. Also to be robust against serious problems, we had to make the script run as a service that restarts itself. To name a few of these problems: script failure, internet connection failure and electric outage.

5.2 API Challenges

Selecting an appropriate weather API proved non-trivial. Initial attempts using Météo-France and OpenWeatherMap were abandoned due to overly complex formats and restrictive pricing plans. While Open-Meteo offered suitable data, its rate limits posed a significant challenge.

Table 6: Open-Meteo Free Plan API Rate Limits

Constraint	Limit
Commercial use	Not permitted
Requests per minute	600
Requests per hour	5,000
Requests per day	10,000
Requests per month	300,000

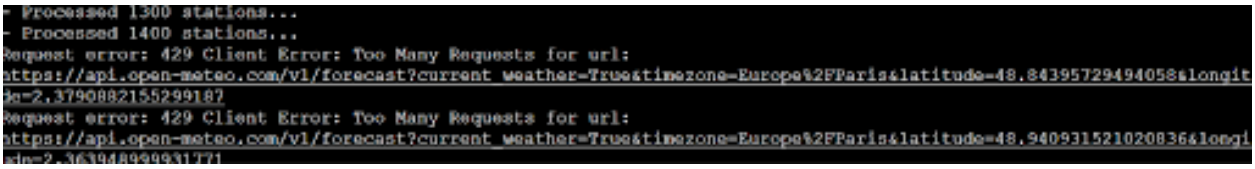


Figure 4: Request Error

After exceeding the request quota during early experiments, we redesigned our collection strategy to drastically reduce the number of API calls. Instead of querying weather data for each station, we selected representative weather points at the commune or arrondissement level. Each Vélib' station was then assigned to its nearest weather point based on geographic distance. This way, the number of requests was reduced from 1500 to 86(20 arrondissements and 66 communes) per 15-20 minutes.

6 Station-Level Feature Extraction

To capture persistent behavioral differences across stations, we constructed a station-level feature table computed exclusively from the training split to avoid information leakage. The goal of this representation is to summarize typical usage patterns and weather sensitivity for each station using robust, interpretable statistics.

6.1 Usage Normalization

Because stations differ substantially in capacity, raw bike counts are not directly comparable across stations. We therefore normalized station usage by station capacity. For each observation, we define the usage ratio as

$$u(t) = \mathrm{clip}\left(\frac{b(t)}{c},\, 0,\, 1\right), \tag{3}$$

where $b(t)$ is the number of available bikes at time t, c is the station capacity, and $\mathrm{clip}(\cdot)$ bounds the ratio to $[0, 1]$.

6.2 Extracted Station Features

Using the normalized usage ratio $u(t)$, we computed the following station-level features on the training set:

- **Basic usage statistics:** mean, standard deviation, minimum, and maximum of $u(t)$.
- **Temporal behavior:** mean usage on weekdays vs. weekends, and during day vs. night, using the dataset indicators `is_weekend` and `is_day`.
- **Daily amplitude:** difference between the maximum and minimum of the average usage profile across hours of the day.
- **Weather correlations:** Pearson correlations between $u(t)$ and each weather variable (temperature, wind speed, wind direction, and weather code), providing a compact measure of weather sensitivity.
- **Capacity:** the most recently observed station capacity value.

Any stations producing undefined statistics (e.g., NaN correlations) were removed from the feature table to ensure numerical stability in downstream clustering.

7 Data Processing and Integration

7.1 Preprocessing Pipeline

Raw data collected from the Vélib' and Open-Meteo APIs cannot be directly used for modeling due to inconsistencies, missing values, and anomalies as we are going to mention later. We therefore implemented a multi-stage preprocessing pipeline, summarized below and implemented in the notebook `prepare_data.ipynb`.

7.1.1 Pipeline Overview

The preprocessing pipeline consists of the following steps:

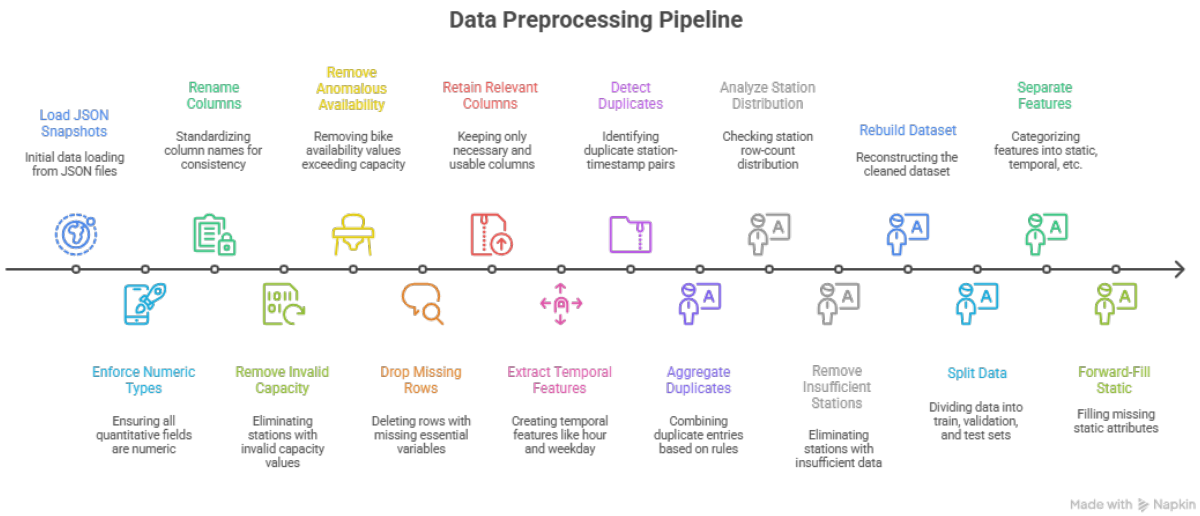


Figure 5: Pipeline

Here is a graph of evaluation in row and station numbers during some of these steps:

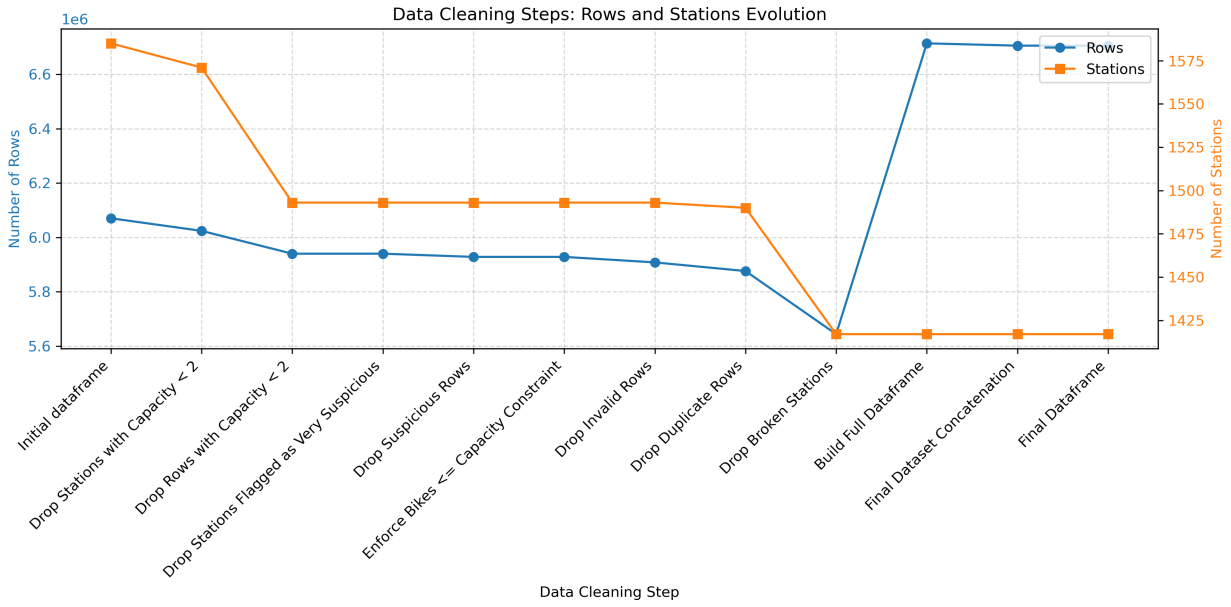


Figure 6: Pipeline Statistics

7.2 Temporal Alignment

Vélib' and weather data are aligned using rounded timestamps at fixed 15-minute intervals. This ensures temporal consistency across sources and facilitates aggregation.

7.3 Spatial Approximation

Distances between stations and weather points are computed using the Euclidean distance. Given the limited spatial extent of the Île-de-France region, using haversine distance for this was too comlex and eudlidian distance approximation was deemed sufficient, despite the Earth's curvature.

7.4 Making Hour Feature Usable

Having the 0-23 hour system for us humans is easy to understand, but because it is not continuous for computers(it drops from 23 to 0). To make it continuous, we used the cosine and sine of the hour using the formula the formulas below. They make sense when used in a couple and with this technique, we can represent the time continuously.

Given an hour value $h \in \{0, 1, \ldots, 23\}$, we define the transformed features as

$$\text{sin_hour} = \sin\left(\frac{2\pi h}{24}\right) \tag{4}$$

$$\text{cos_hour} = \cos\left(\frac{2\pi h}{24}\right) \tag{5}$$

7.5 Cleaning and Aggregation

We have collected the Vélib' and weather data in a script, where we have also merged the collected data at the same time. Even though we have set the data collection to collect the data each 15-20 minutes, sometimes due to connection and hardware limitations the collected data had some problems. Several data quality issues were identified:

- Failed API requests during data collection.
- Missing or duplicated timestamps.
- Inexplicable bike and capacity data.
- Stations with intermittent outages.

7.5.1 Failed API Requests During Data Collection

Open-Meteo API has a rate limit and as mentioned in earlier sections, we have changed our data collection script to not pass this limit and to be robust as possible. But even so, some of the requests still returned errors. While this is not necessarily a rate limit problem, this still caused some empty values in the dataset. With a weather point being connected to numerous Vélib' stations, one of the weather points not existing meant not having weather values for these stations for that specific request. While this was not possible to fix it directly in the script, we have solved this by later filling the data using aggregation.

7.5.2 Missing or Duplicated Timestamps

Fixing the timestamps to 15-minute marks proved to have introduce possible problems in the dataset. After analyzing, we have seen that number of (station, timestamp) couples seemed to range between 0 and 4.

Table 7: Station Availability Records

Station Code	Date	Hour	Minute	Present
14111	2025-11-07	3	0	1
14111	2025-11-07	3	15	1
14111	2025-11-07	3	30	0
14111	2025-11-07	3	45	4
14111	2025-11-07	4	0	4
14111	2025-11-07	4	15	2
14111	2025-11-07	4	30	1
14111	2025-11-07	4	45	1

We found the solution in merging the existing duplicate couples and aggregating the missing ones with forward fill. To avoid data-leak, we filled the fields after splitting the data.

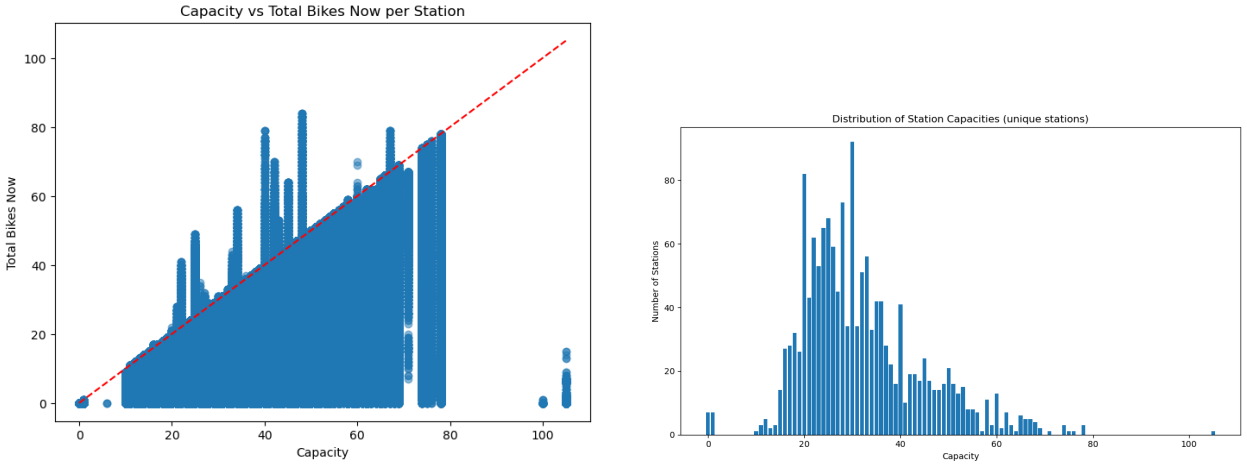
Table 8: Aggregation Rules Applied to Duplicate Station-Timestamp Features

Feature	Aggregation Method
Latitude	First value
Longitude	First value
Station capacity	First value
Total bikes available	Mean
Total bikes taken	Mean
Temperature	Mean
Wind speed	Mean
Wind direction	Mean
Weather code	First value
Daytime indicator	First value
Hour of day	First value
Day of week	First value
Weekend indicator	First value
Sine of hour	First value
Cosine of hour	First value

To remove duplicate observations, we grouped the dataset by `station_code` and `timestamp` and aggregated all remaining variables according to the rules summarized in Table 8. Static station attributes (like location and capacity) were retained using their first observed values, while dynamic usage and weather-related variables were aggregated using the mean. This process ensures a single, consistent record per station and timestamp.

7.6 Inexplicable Bike and Capacity Data

Because of possible errors from the Vélib servers, we have detected anomalies in some bike and capacity data. These could harm the dataset badly by causing a lot of noise as these were not only a few rows but several stations and numerous rows.



(a) Capacity vs. total bikes available

(b) Distribution of station capacities

Figure 7: Analysis of station capacity characteristics

Figure 7(a) shows the relationship between station capacity and bike availability, while Figure 7(b) presents the distribution of station capacities.

We have detected 14 stations having consistently 0 or 1 capacity, which is not normal and not usable for our dataset. So we just deleted those stations and their rows.

We later analyzed the bike numbers, which did not match with the capacity numbers. After checking the percentage of these suspicious rows to the total row number of the station, we have decided to drop the stations with suspicious

data more than 10% of the rows and just dropping the suspicious rows as most of them were under 1% of the total row number of the station. 9 stations were dropped and corrupt rows of 22 stations were removed.

Table 9: Stations with High Proportions of Suspicious Observations

Station Code	Total Rows	Suspicious Rows	Suspicious Ratio
21302	3970	2319	0.584
42004	4014	2165	0.539
13123	4014	1713	0.427
15056	2543	746	0.293
12105	2579	569	0.221
12127	3464	760	0.219
21021	4014	743	0.185
33019	3985	661	0.166
15058	3991	601	0.151
12010	3966	352	0.089

7.6.1 Stations with Intermittent Outages

While analyzing the dataset, we have seen that some stations have very low row counts, while most were missing up to 15 rows from the stations that had the most rows.

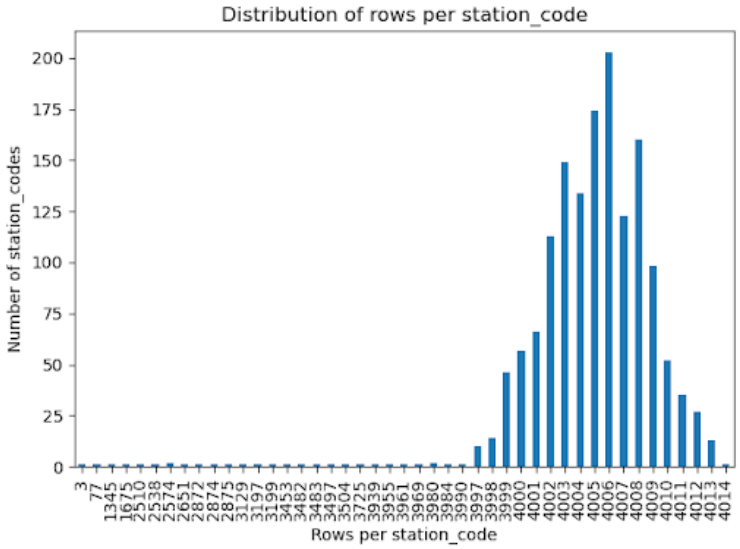


Figure 8: Station Row Count Histogram

We have fixed this by applying several rules to detect such stations and removing them, such as:

- Station row count being lower than 5% of the stations
- For available bike and dock number:
  - Standart deviation being lower than 0.1 or NaN.
  - Mean being less than 0.5.

We had this result after filtering:

- 67 stations removed by row count
- 11 stations removed by bikes_now(bikes) stats
- 9 stations removed by bikes_taken(docks) stats

- Stations to remove: 73
- Stations to keep: 1417

Table 10: Station Filtering Summary

Filtering Criterion	Number of Stations
Removed due to row count	67
Removed due to bikes_now (bikes) statistics	11
Removed due to bikes_taken (docks) statistics	9
Total stations removed	73
Stations retained	1417

8 Feature Engineering

From the raw Vélib' and weather data, several features were constructed to capture temporal and contextual patterns affecting bike availability.

Table 11: Extracted Features for Bike Availability Prediction

Feature Category	Feature Name
Basic Usage Features	Usage ratio
	Mean usage
	Standard deviation of usage
	Minimum usage
	Maximum usage
Temporal Features	Weekday mean
	Weekend mean
	Day usage
	Night usage
	Daily amplitude
Weather Correlation Features	Correlation with temperature
	Correlation with wind speed
	Correlation with wind direction
	Correlation with weather code

These features allow models to learn daily commute cycles and weekly usage patterns along with correlating weather features.

9 Dataset Description

The final dataset contains approximately 6.7 million rows and 21 columns, of which 11 are used as predictive features. Each row corresponds to a unique (station, timestamp) pair.

Table 12: Description of Dataset Columns

Column	Type	Description
timestamp	datetime	Timestamp of the observation
station_code	integer	Unique identifier of the Vélib' station
latitude	float	Latitude of the station
longitude	float	Longitude of the station
capacity	float	Total number of docks at the station
total_bikes_now	float	Number of bikes available at the current timestamp (target variable)
total_bikes_taken	float	Number of docks available
temperature	float	Air temperature at 2 meters above ground
wind_speed	float	Wind speed at 10 meters above ground
wind_direction	float	Wind direction at 10 meters above ground
weather_code	float	Weather condition code defined by the WMO
is_day	float	Indicator of daylight (1 if day, 0 if night)
hour	integer	Hour of day (0–23)
day_of_week	integer	Day of week (0=Monday, 6=Sunday)
is_weekend	integer	Indicator for weekend days
sin_hour	float	Sine encoding of hour-of-day
cos_hour	float	Cosine encoding of hour-of-day
total_bikes_lag1	float	Bikes available 15 minutes earlier
total_bikes_lag2	float	Bikes available 30 minutes earlier
split	integer	Data split indicator (train:0 / validation:1 / test:2)

The data are split chronologically into training (70%), validation (15%), and test (15%) sets to prevent temporal leakage.

10 Exploratory Analysis

Exploratory analysis revealed strong temporal patterns, such as daily and weekly usage cycles. In contrast, weather variables exhibited relatively weak linear correlations with bike availability. Lagged bike availability emerged as the most informative predictor, highlighting the dominance of habitual mobility patterns.

11 Station Clustering

After preprocessing, the dataset contains approximately 1,400 Vélib' stations distributed across the Île-de-France region. These stations exhibit heterogeneous usage patterns due to differences in location, demand, and surrounding urban context. Training a separate predictive model for each station would be computationally inefficient and statistically unstable for sparsely observed stations.

To address this heterogeneity while maintaining tractable modeling complexity, we group stations into clusters based on long-term station characteristics. We evaluate two clustering strategies: one using both behavioral and weather-related features, and one using behavioral features only. This comparison allows us to assess the impact of weather information on station differentiation and cluster quality.

11.1 Station Feature Sets

Clustering is performed using station-level features extracted exclusively from the training split, ensuring that no information leakage occurs. These features summarize typical station behavior and long-term sensitivity to weather conditions.

Two feature groups are considered:

Table 13: Groups of Station-Level Features Used for Clustering

Behavioral Features	**Weather Sensitivity Features**
Mean usage ratio	Day usage ratio
Mean weekend usage	Night usage ratio
Mean weekday usage	Correlation with temperature
Standard deviation of usage	Correlation with wind speed
Daily amplitude	Correlation with wind direction
	Correlation with weather code

11.2 Choice of the Number of Clusters

Stations are clustered using K-Means applied to standardized feature vectors. The number of clusters is selected using the elbow method and the silhouette score.

11.2.1 Comparison of Cluster Selection Criteria

Figure 11 compares the elbow curves obtained for the two feature configurations.

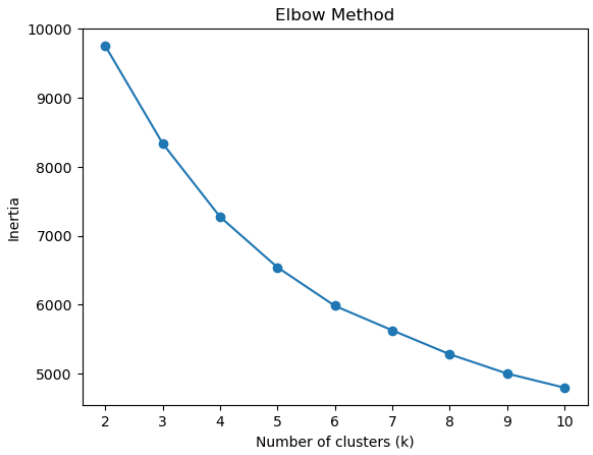


Figure 9: Elbow method (full feature set)

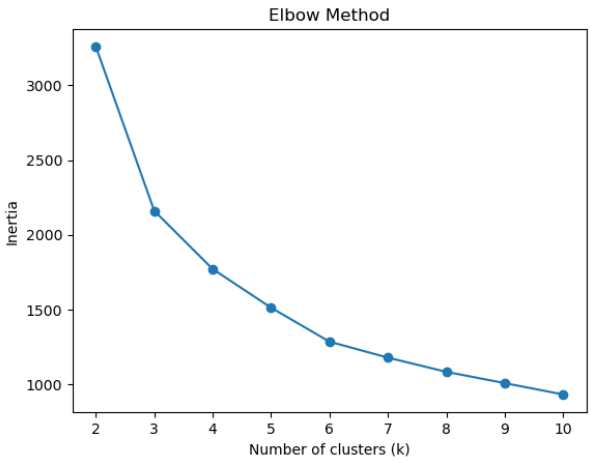


Figure 10: Elbow method (behavioral features only)

Figure 11: Comparison of elbow curves for the two clustering configurations.

In both cases, the elbow method does not reveal a sharp inflection point, suggesting that station characteristics evolve along a continuum rather than forming clearly separated groups. Silhouette scores therefore provide a more discriminative criterion and are used to select the final number of clusters.

11.2.2 Clustering Using the Full Feature Set

Silhouette scores obtained using the full feature set are reported below:

- $K = 2$: silhouette score = 0.338
- $K = 3$: silhouette score = 0.309
- $K = 4$: silhouette score = 0.298
- $K = 5$: silhouette score = 0.296
- $K = 6$: silhouette score = 0.212
- $K = 7$: silhouette score = 0.205

The highest silhouette score is achieved for $K = 2$, which is therefore retained. This partitions the stations into two clusters of comparable size (682 and 734 stations).

11.2.3 Clustering Using Behavioral Features Only

Silhouette scores obtained using behavioral features only are consistently higher:

- $K = 2$: silhouette score = 0.474
- $K = 3$: silhouette score = 0.458
- $K = 4$: silhouette score = 0.359
- $K = 5$: silhouette score = 0.332
- $K = 6$: silhouette score = 0.325
- $K = 7$: silhouette score = 0.315

The substantial improvement in silhouette scores confirms that behavioral features alone provide a more coherent clustering structure. As in the full feature case, $K = 2$ is retained.

11.3 Feature Space Visualization

After selecting the number of clusters, Principal Component Analysis (PCA) is used to visualize the resulting partitions. Stations are projected onto the first two principal components and colored according to their assigned cluster.

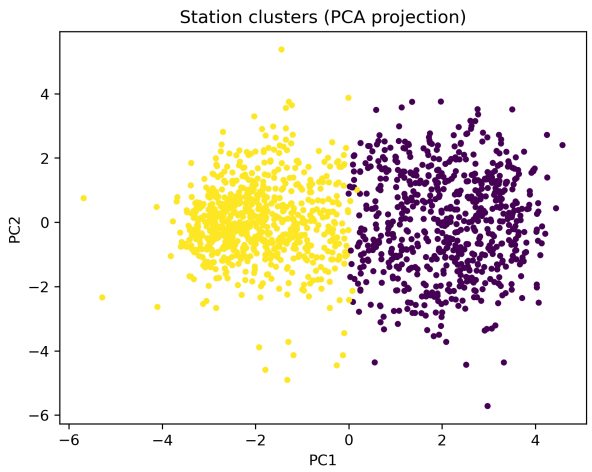


Figure 12: PCA projection with $K = 2$ (full feature set)

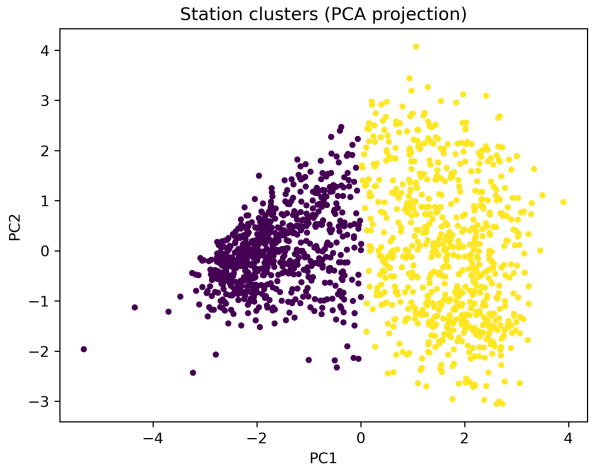


Figure 13: PCA projection with $K = 2$ (behavioral features only)

When using the full feature set, the two clusters exhibit substantial overlap in the PCA projection, indicating weak geometric separation. In contrast, the behavioral-only configuration yields a clearer partition, consistent with its higher silhouette scores.

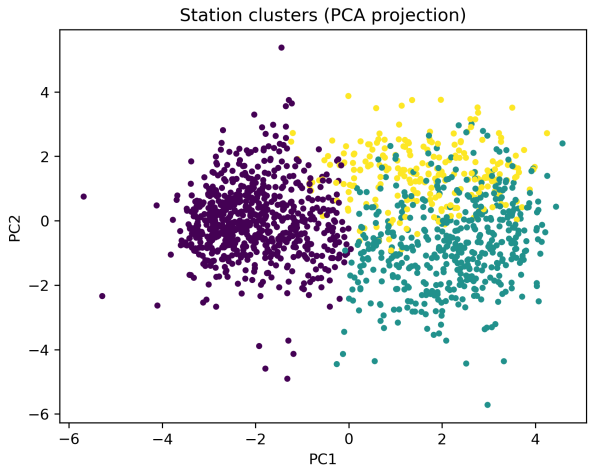


Figure 14: PCA projection with $K = 3$ (full feature set)

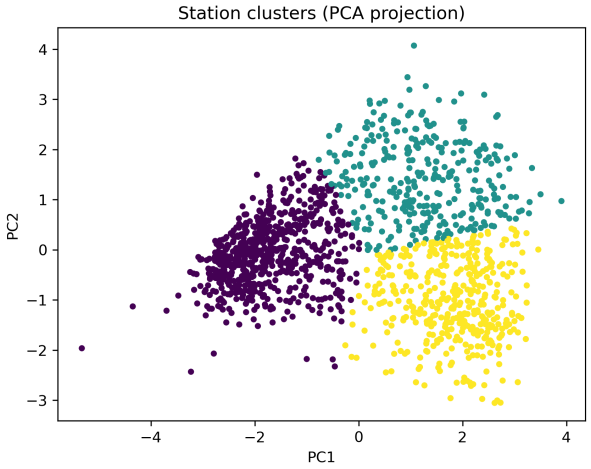


Figure 15: PCA projection with $K = 3$ (behavioral features only)

For larger values of K (e.g., $K = 3$), PCA projections reveal that weather-related features tend to drive overlapping clusters, while behavioral features preserve more distinct groupings. This observation further supports the dominance of usage dynamics in station differentiation.

11.4 Cluster Interpretation and Comparison

Tables 14 and 15 summarize the clusters obtained using the full feature set.

Table 14: Mean Usage and Temporal Features per Cluster (Full Feature Set)

Cluster	Mean	Weekend	Weekday	Std	Amplitude
0	0.556	0.560	0.554	0.248	0.327
1	0.244	0.248	0.242	0.179	0.236

Table 15: Mean Weather-Related Features per Cluster (Full Feature Set)

Cluster	Day	Night	Corr. Temp	Corr. Wind Spd	Corr. Wind Dir	Corr. W.Code
0	0.568	0.548	-0.020	0.040	0.021	0.023
1	0.216	0.260	-0.056	-0.073	-0.011	-0.002

Cluster 0 corresponds to high-usage stations with stronger daily dynamics, while Cluster 1 groups lower-usage stations with more stable behavior. Weather-related features, particularly day and night usage ratios, exhibit strong internal correlation within clusters, indicating consistent diurnal patterns. However, these variables do not contribute to improved cluster separation, as reflected by lower silhouette scores and increased overlap in PCA projections when weather features are included.

Weather-related variables were therefore excluded from the final clustering used for modeling, in order to focus on behavioral differentiation and to assess explicitly whether weather information provides additional explanatory value.

Using behavioral features only yields the cluster statistics reported in Table 16, with a clearer separation between high-usage and low-usage stations.

Table 16: Mean Behavioral Features per Cluster (Behavioral Only)

Cluster	Mean Usage	Weekend	Weekday	Std	Amplitude
0	0.240	0.245	0.238	0.177	0.229
1	0.550	0.553	0.548	0.249	0.332

Overall, this comparison shows that behavioral features dominate station differentiation. Including weather-related variables reduces cluster coherence without providing additional explanatory power. Consequently, clusters derived from behavioral features are retained for training cluster-specific prediction models in subsequent sections.

12 Modeling Approach

This section describes the supervised learning setup used to predict Vélib' bike availability. We detail the problem formulation, the per-cluster modeling strategy, the input features and target variable, and the regression models evaluated. The focus is on methodological design rather than implementation details.

12.1 Problem Setup

12.1.1 Per-Cluster Modeling Strategy

As established in Section 11, Vélib' stations exhibit heterogeneous long-term usage patterns. To account for this heterogeneity while avoiding station-specific models, we adopt a *per-cluster modeling* strategy.

Stations are first grouped into clusters based on long-term behavioral features. For each cluster, a separate regression model is trained using only observations from stations belonging to that cluster. This approach allows models to specialize to homogeneous usage regimes while maintaining sufficient data volume for robust training.

12.1.2 Input Features and Target Variable

The prediction task is formulated as a supervised regression problem. The target variable is the number of available bikes at a given station and timestamp.

Input features include:

- **Lagged availability features**, capturing short-term temporal dependence (e.g., bike availability 15 and 30 minutes earlier).
- **Contextual features**, such as capacity.
- **Temporal features**, including cyclic encodings of hour of day and indicators for weekdays and weekends.
- **Weather variables**, such as temperature and wind speed, used to assess their marginal contribution beyond behavioral dynamics.

Table 17: Main Features Used for Prediction

Feature	Interpretation
sin_hour, cos_hour	Encodes time of day continuously
is_weekend	Differentiates workdays from weekends
capacity	Number of docks at the station
temperature	Air temperature at station area 2 meters above ground
wind_speed	Wind speed 10 meters above ground
wind_direction	Wind direction 10 meters above ground
weather_code	Designed by the WMO, gives the weather situation (sunny, rainy, etc.)
is_day	Existence of daylight
total_bikes_lag1	Bikes available 15 minutes earlier
total_bikes_lag2	Bikes available 30 minutes earlier

All features are constructed to respect temporal causality, and no information from future timestamps is used.

12.2 Target Variable

The target variable is the number of available bikes at a station at a given timestamp. This quantity is bounded between zero and the station capacity and is treated as a continuous regression target.

12.2.1 Train / Validation / Test Split

To avoid temporal leakage, data are split chronologically into three disjoint sets:

- **Training set**: 70% of the earliest observations
- **Validation set**: 15% of subsequent observations
- **Test set**: 15% of the most recent observations

The split is applied independently within each cluster. Model selection and hyperparameter evaluation are performed on the validation set, while final performance is reported on the test set only.

12.3 Models Used

We evaluate three regression model families with increasing expressive power. This progression allows us to compare linear baselines against non-linear models capable of capturing complex interactions.

12.3.1 Linear Regression

Linear Regression serves as a simple and interpretable baseline. It assumes a linear relationship between input features and bike availability. Despite its limited capacity, it provides a useful reference point for assessing the added value of more complex models.

12.3.2 SGD Regressor

The SGD Regressor is a linear model trained via stochastic gradient descent. It is well-suited for large-scale datasets and allows efficient optimization when features are standardized. Compared to ordinary linear regression, it offers greater flexibility in handling correlated features and large training sets.

12.3.3 Random Forest Regressor

The Random Forest Regressor is a non-linear ensemble model based on decision trees. By averaging predictions over multiple trees, it can capture complex feature interactions and non-linear effects while remaining relatively robust to noise. This model represents the most expressive approach evaluated in this study.

13 Evaluation Metrics

Model performance is evaluated using three complementary regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Together, these metrics provide a balanced assessment of prediction accuracy, error magnitude, and explanatory power.

13.1 Mean Absolute Error (MAE)

The Mean Absolute Error measures the average absolute deviation between predicted and observed bike availability values:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{6}$$

MAE is expressed in the same units as the target variable and is robust to outliers. It provides an intuitive interpretation of typical prediction error.

13.2 Root Mean Squared Error (RMSE)

The Root Mean Squared Error penalizes larger prediction errors more strongly:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{7}$$

RMSE is particularly relevant for operational settings, where large errors in bike availability prediction may have a disproportionate impact on user experience.

13.3 Coefficient of Determination (R^2)

The coefficient of determination measures the proportion of variance in the target variable explained by the model:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{8}$$

An R^2 value close to 1 indicates strong predictive performance, while negative values signal performance worse than a constant baseline predictor.

14 Results and Analysis

This section analyzes the predictive performance of the evaluated models across clusters, feature configurations, and model families. Rather than reporting isolated numerical scores, we focus on systematic trends that reveal the relative importance of temporal dependence, station heterogeneity, and exogenous information.

14.1 Model Comparison

14.1.1 Performance per Model Family

Across all configurations, model performance is dominated by the presence of lagged bike availability features. When lag features are excluded, all models exhibit limited explanatory power, with R^2 values remaining below 0.41 even for Random Forest regressors. Performance differences between model families remain modest, indicating that contextual and weather variables alone are insufficient to capture short-term availability dynamics.

Table 18 reports performance metrics for models trained without lag features using the full feature set.

Table 18: Model Performance without Lag Features (All Features)

Cluster	Model	R^2	MAE	RMSE
0	Random Forest	0.383	7.47	9.41
0	Linear Regression	0.354	7.68	9.64
0	SGD Regressor	0.353	7.68	9.64
1	Random Forest	0.181	4.84	6.43
1	SGD Regressor	0.163	4.85	6.50
1	Linear Regression	0.163	4.86	6.50

When lag features are included, predictive performance improves dramatically for all model families. Linear Regression and SGD Regressor yield nearly identical results across clusters, confirming that the optimization method has little impact relative to feature selection. Random Forest models consistently achieve the best performance, though the improvement over linear models remains moderate once lagged availability is included.

Table 19 summarizes results obtained with lag features and the full feature set.

Table 19: Model Performance with Lag Features (All Features)

Cluster	Model	R^2	MAE	RMSE
0	Random Forest	0.970	1.17	2.08
0	Linear Regression	0.968	1.25	2.14
0	SGD Regressor	0.968	1.25	2.15
1	Random Forest	0.944	0.95	1.68
1	Linear Regression	0.940	1.03	1.74
1	SGD Regressor	0.940	1.04	1.74

The contrast between Tables 18 and 19 highlights the decisive role of short-term temporal information. The inclusion of bike availability lags increases R^2 by more than 0.6 in all cases, dwarfing the impact of model choice or additional contextual features.

Figures 16 and 17 illustrate the distribution of R^2 values across model families for both clusters.

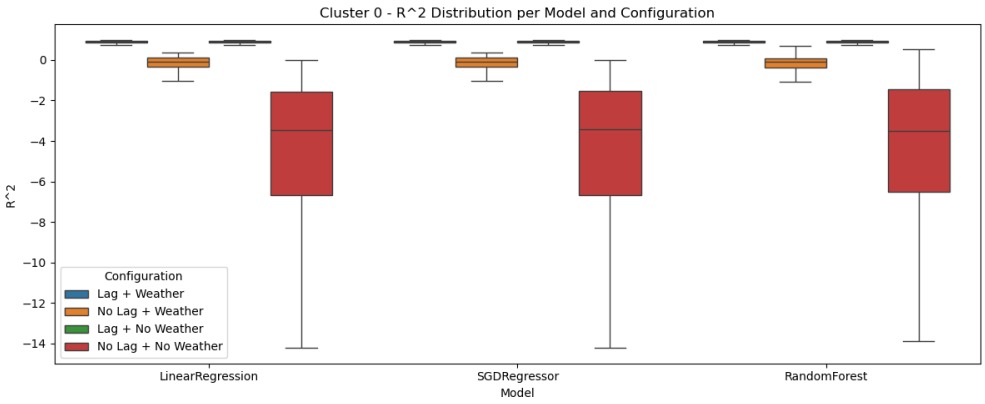


Figure 16: R^2 distribution by model family for Cluster 0

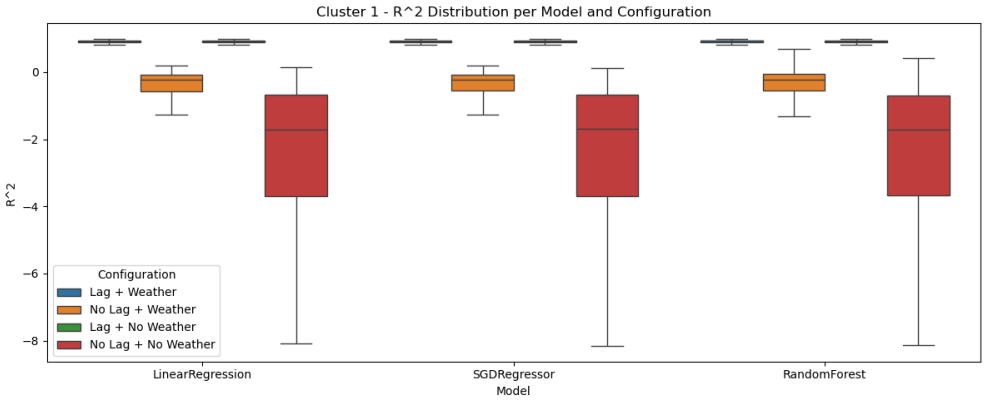


Figure 17: R^2 distribution by model family for Cluster 1

14.1.2 Performance per Cluster

Clusters exhibit distinct predictability profiles. When lag features are included, Cluster 1 consistently achieves higher R^2 values (up to 0.97) and lower RMSE values than Cluster 0. This suggests that low-usage stations exhibit smoother and more predictable short-term dynamics.

In contrast, when lag features are excluded, Cluster 0 outperforms Cluster 1. This reversal indicates that high-usage stations benefit more from contextual and temporal features alone, whereas low-usage stations are particularly difficult to predict without short-term historical information.

Figures 18 and 19 summarize RMSE distributions.

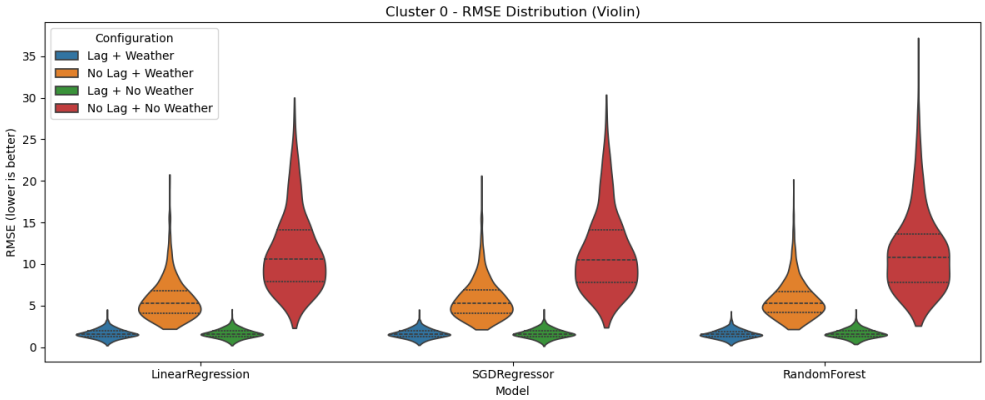


Figure 18: RMSE distribution for Cluster 0

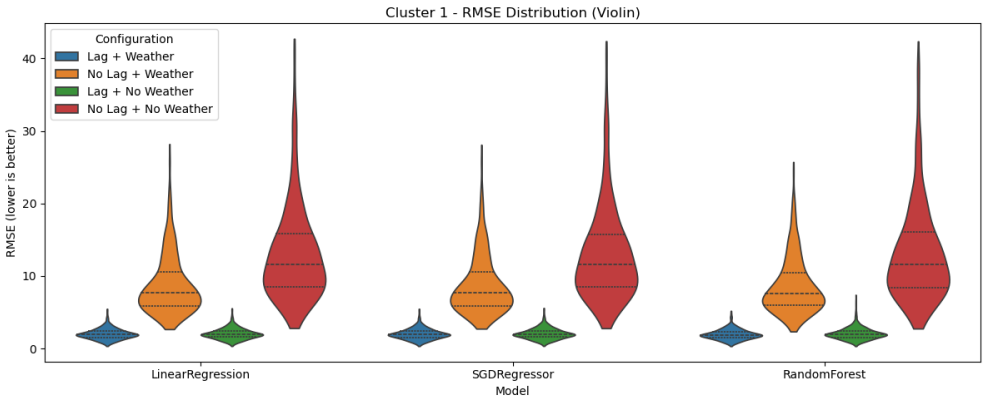


Figure 19: RMSE distribution for Cluster 1

14.2 Success and Failure Cases

14.2.1 Stations with Strong Predictive Performance

Station-level evaluation reveals that a large fraction of stations achieve very high predictive accuracy when lagged availability features are included. In these cases, R^2 values frequently exceed 0.9 across all model families, indicating that short-term autoregressive dynamics dominate bike availability prediction.

For such stations, the inclusion of weather variables does not materially affect performance, suggesting that recent historical availability already captures most of the relevant information at the considered prediction horizon.

14.2.2 Stations with Lower Predictive Performance

Despite strong average performance, station-level results exhibit substantial heterogeneity. In particular, when lagged availability features are excluded, a non-negligible subset of stations attains near-zero or negative R^2 values, corresponding to performance worse than a constant baseline predictor.

These observations indicate that, for certain stations, contextual and weather-related features alone are insufficient to recover short-term availability dynamics. While this work does not explicitly model the underlying causes at the station level, such behavior is consistent with stations exhibiting weak temporal continuity, irregular usage patterns, or unobserved operational effects.

Importantly, the inclusion of weather variables does not systematically improve performance for these lower-performing cases, reinforcing the conclusion that short-term historical information is the primary driver of predictability in this setting.

This highlights an inherent limitation of short-horizon regression models when temporal dependence is weak or unavailable.

14.3 Feature Impact Analysis

Feature importance analyses confirm that lagged bike availability variables overwhelmingly dominate prediction across all clusters and model families. In particular, Random Forest models assign the majority of feature importance to availability at $t-15$ and $t-30$ minutes, often concentrating nearly all importance mass on these lagged variables. This behavior is consistent with the large performance gap observed between models trained with and without lag features.

Linear Regression and SGD Regressor models distribute importance a little more evenly across features, assigning non-negligible weights to temporal and contextual variables in addition to bike lags. However, even in these linear models, lagged availability remains the most influential predictor whenever it is included.

Figures 20 and 21 present feature importance rankings under lagged configurations, confirming the dominance of short-term temporal dependence.

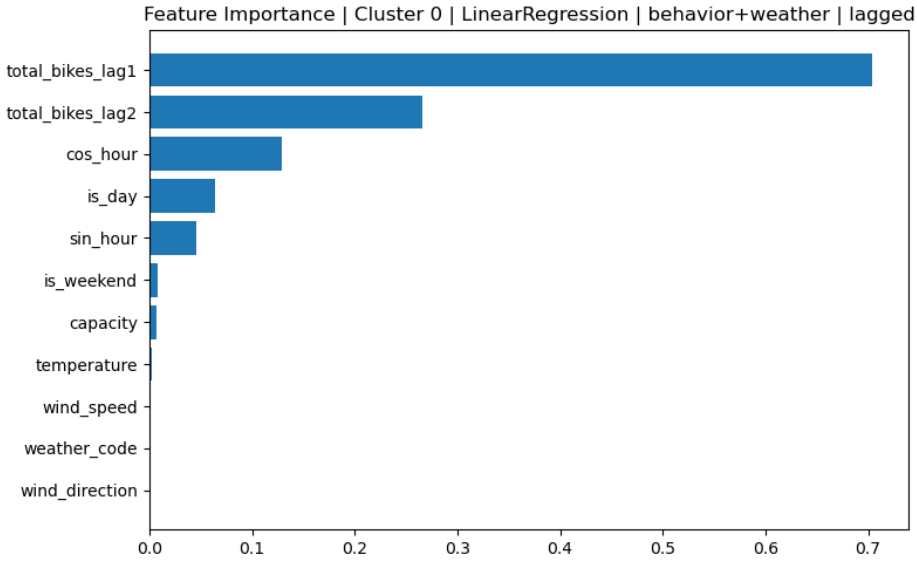


Figure 20: Feature importance for Linear Regression (Cluster 0, all features + lag)

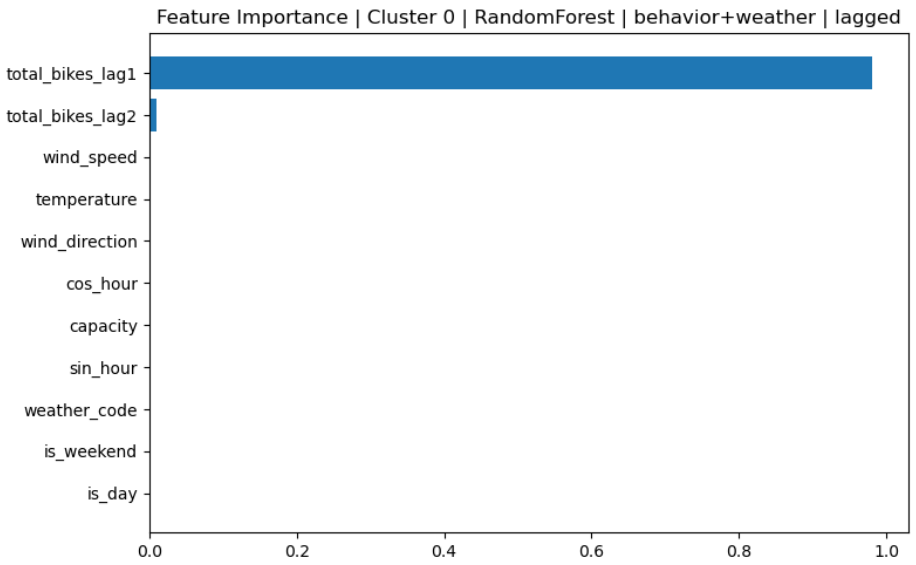


Figure 21: Feature importance for Random Forest(Cluster 0, all features + lag)

When lag features are included, temporal encodings of hour-of-day contribute secondarily by capturing daily usage cycles, while contextual features such as capacity lose relative importance. Weather variables consistently exhibit low importance and unstable contributions across configurations, regardless of the model family or cluster.

Figures 22 and 23 illustrate representative feature importance rankings for SGD Regressor and Random Forest without lag features. In this setting, contextual features such as station capacity dominate the importance scores, reflecting the lack of short-term temporal information.



Figure 22: Feature importance for SGD Regressor (Cluster 1, no lag features)

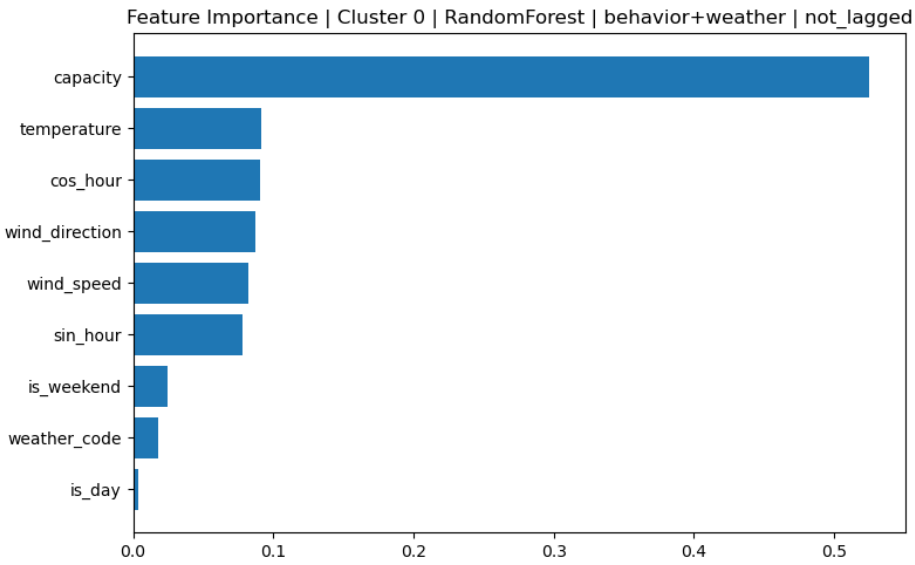


Figure 23: Feature importance for Random Forest (Cluster 0, no lag features)

Overall, these results demonstrate that Vélib' bike availability is governed primarily by short-term temporal dependence, with limited marginal benefit from weather-related variables at the considered prediction horizon.

15 Discussion

This section critically examines the strengths and limitations of the proposed modeling approach, with particular attention to its assumptions, robustness, and generalization potential.

15.1 What Worked Well

The most significant success of this study lies in the ability of simple regression models to achieve very high predictive performance when short-term temporal information is available. The inclusion of lagged bike availability features enables all evaluated models to capture the dominant autoregressive structure of Vélib' usage dynamics, leading to R^2 values exceeding 0.9 across most stations.

The clustering-based modeling strategy also proved effective. By grouping stations according to long-term behavioral characteristics, the approach balances model specialization with statistical stability, avoiding the drawbacks of both global and station-specific models.

Finally, the consistency of results across linear and non-linear models indicates that the core predictive signal is robust and does not rely on excessive model complexity.

15.2 Where the Approach Struggled

Despite strong average performance, the approach struggles in the absence of lagged availability information. Models trained without bike lags exhibit low explanatory power, particularly for low-usage stations, highlighting a strong dependence on historical continuity.

Additionally, a subset of stations remains difficult to predict even under lagged configurations. These failure cases are typically associated with sparse usage, abrupt availability changes, or operational irregularities such as maintenance or rebalancing events that are not explicitly modeled.

15.3 Limitations of Data and Models

Several limitations should be acknowledged. First, the models rely exclusively on historical availability and coarse contextual features, without access to operational data such as bike rebalancing actions, station outages, or special events. These unobserved factors likely contribute to unexplained variance.

Second, weather data are spatially interpolated and temporally coarse relative to station-level dynamics, which may partially explain their limited predictive contribution. However, even under ideal conditions, weather effects appear secondary at the short prediction horizon considered.

Finally, the regression framework assumes stationarity of usage patterns within the training period, which may not hold under long-term structural changes in mobility behavior.

15.4 Generalization Concerns

While the results demonstrate strong predictive performance for the studied dataset, generalization to other cities or longer forecasting horizons is not guaranteed. Vélib' exhibits strong routine-driven usage patterns that may not transfer directly to bike-sharing systems with different urban layouts, user bases, or operational strategies.

Moreover, performance at short horizons does not necessarily imply accuracy at longer horizons, where exogenous factors and structural trends may play a larger role.

16 Ethical and Legal Considerations

All data used are publicly available and anonymized. No personal information is collected. The project complies with GDPR regulations and respects API usage policies. Potential biases may arise from spatial heterogeneity between urban and suburban stations.

17 Conclusion and Future Work

This project investigated the short-term prediction of Vélib' bike availability, a problem of practical relevance for both users and operators of bike-sharing systems. Using large-scale historical data, the study combined feature engineering, station clustering, and supervised regression models to analyze the determinants of short-horizon predictability.

The main findings of this work can be summarized as follows:

- Lagged bike availability is by far the dominant predictor of short-term availability, largely outweighing the contribution of contextual and weather-related features.
- Clustering stations based on long-term behavioral characteristics improves modeling efficiency while preserving predictive performance.
- Weather variables provide limited marginal benefit at short prediction horizons, challenging common assumptions about weather-driven mobility demand.

Several directions for future work naturally follow from these results. Extending the temporal coverage of the dataset would allow the analysis of seasonal effects and long-term behavioral changes. Improved spatial and temporal resolution of weather data could help determine whether more localized environmental signals influence bike availability.

Beyond feature refinement, future models could integrate operational information such as bike rebalancing actions, station outages, or nearby transport disruptions, which are not captured in the current dataset. Finally, advanced spatio-temporal approaches, including recurrent or graph-based neural networks, may offer improved performance for longer forecasting horizons and enable explicit modeling of cross-station dependencies.

Overall, this work highlights the importance of critically evaluating assumed explanatory factors and demonstrates that, for short-term prediction, routine-driven temporal dynamics remain the primary determinant of bike-sharing availability.

References

[1] Vélib' Métropole. Vélib' open data api, 2024. Accessed: 2025-01-10. URL: `https://www.velib-metropole.fr/donnees-open-data-gbfs-du-service-velib-metropole`.

[2] Open-Meteo. Open-meteo weather api, 2024. Accessed: 2025-01-10. URL: `https://open-meteo.com`.